

Data Model Documentation

Overview

The expense management application manages the following core domains:

- Employee: Responsible for submitting and approving expenses, cardinality is 1:N with Expense.
- Expense: Represents the expenses submitted by employees, cardinality is 1:N with Employee and 1:1 with Approval.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee who can submit and approve expenses.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	String	Yes	Max 255 chars	Employee name
email	String	Yes	Max 255 chars, unique	Employee email
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many Expense (1:N): Inverse via Expense . employee_id
- Has many Approval (1:N): Inverse via Approval . approver_id
- Has many Report (1:N): Inverse via Report . employee_id

Validation Rules:

- name must be non-empty
- email must be non-empty and unique across all records

Expense

Purpose: Represents an expense submitted by an employee.

Table/Collection Name: Expense

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
employee_id	UUID	Yes	Foreign key to Employee	Employee who submitted the expense
amount	Decimal	Yes	Min 0	Expense amount
description	String	Yes	Max 255 chars	Expense description
status	String	Yes	'PENDING', 'APPROVED', 'REJECTED'	Expense status
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee . id
- Has one Approval (1:1): Inverse via Approval . expense_id

Validation Rules:

- amount must be greater than or equal to 0

- status must be one of 'PENDING', 'APPROVED', 'REJECTED'

Approval

Purpose: Represents the approval process for an expense.

Table/Collection Name: Approval

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
expense_id	UUID	Yes	Foreign key to Expense	Expense being approved
approver_id	UUID	Yes	Foreign key to Employee	Employee who approved the expense
status	String	Yes	'PENDING', 'APPROVED', 'REJECTED'	Approval status
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Expense (N:1): Foreign key expense_id references Expense.id
- Belongs to Employee (N:1): Foreign key approver_id references Employee.id

Validation Rules:

- status must be one of 'PENDING', 'APPROVED', 'REJECTED'

Report

Purpose: Represents a report generated by an employee.

Table/Collection Name: Report

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier

Field Name	Type	Required	Constraints	Description
employee_id	UUID	Yes	Foreign key to Employee	Employee who generated the report
report_date	Date	Yes		Date of the report
total_amount	Decimal	Yes		Total expense amount
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee.id

Validation Rules:

- report_date must be a valid date

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
Employee	id	Expense	employee_id	1:N	CASCADE
Employee	id	Approval	approver_id	1:N	CASCADE
Employee	id	Report	employee_id	1:N	CASCADE
Expense	id	Approval	expense_id	1:1	CASCADE

Unique Constraints

- Employee(email): Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- Employee.email must be unique: Violation results in a 400 error with a message "Email already exists."

- `Expense.amount` must be greater than or equal to 0: Violation results in a 400 error with a message "Amount must be greater than or equal to 0."
- `Approval.status` must be one of 'PENDING', 'APPROVED', 'REJECTED': Violation results in a 400 error with a message "Invalid approval status."

Format Rules:

- `Employee.email`: Must match the regex
`^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`
- `Expense.date`: Must be in the format YYYY-MM-DD

Data Lifecycle

Creation

- A new `Employee` is created when a user signs up.
- Default status for `Expense` is `PENDING`.
- `created_by` is automatically set to the current user.

Updates

- `Employee.email` can only be updated by the employee themselves.
- `updated_at` is automatically updated on any modification.

Archival / Soft Delete

- Records are not archived but can be marked as deleted by setting the status to 'DELETED'.

Indexes

- Primary key: `id`
- Foreign keys: `employee_id`, `expense_id`, `approver_id`
- Search fields: `name`, `email`, `description`

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2023-10-01T10:30:00Z",
  "updated_at": "2023-10-01T10:30:00Z"
}
```

Expense

```
{
  "id": "550e8400-e29b-41d4-a716-446655440001",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "amount": 100.00,
  "description": "Lunch",
  "status": "PENDING",
  "created_at": "2023-10-01T10:30:00Z",
  "updated_at": "2023-10-01T10:30:00Z"
}
```

Approval

```
{
  "id": "550e8400-e29b-41d4-a716-446655440002",
  "expense_id": "550e8400-e29b-41d4-a716-446655440001",
  "approver_id": "550e8400-e29b-41d4-a716-446655440003",
  "status": "APPROVED",
  "created_at": "2023-10-02T10:30:00Z",
  "updated_at": "2023-10-02T10:30:00Z"
}
```

Report

```
{
  "id": "550e8400-e29b-41d4-a716-446655440004",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "report_date": "2023-10-01",
  "total_amount": 1000.00,
  "created_at": "2023-10-03T10:30:00Z",
  "updated_at": "2023-10-03T10:30:00Z"
}
```